Chapter 4: Method of Frobenius

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Outline

1. The guitar string and the drum head
2. When will the method of Frobenius work?
3. Second solutions

Where we are going

Separation of variables turned our PDEs into linear, second-order, homogeneous ordinary differential equations. Two of them recur:

- the **guitar string** gave the harmonic oscillator equation

$$\frac{d^2y}{dx^2} + k^2 y = 0,$$

- the **drum** gave **Bessel's equation**

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (k^2x^2 - m^2)\, y = 0.$$

This chapter develops one general method for the whole class

$$\frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)\, y = 0.$$

Second-order means we have a second derivative of y. *Linear* means P and Q depend only on x, not on y. *Homogeneous* means the right-hand side is zero. Just as analytic and meromorphic functions had Taylor and Laurent representations, we will solve these ODEs with **series**. That technique is the *method of Frobenius*.

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Guitar string: the harmonic oscillator by Taylor series

Solve the harmonic oscillator (here k has been rescaled into x, i.e. $kx \mapsto x$):

$$\frac{d^2y}{dx^2} + y = 0, \qquad y(0) = 0, \quad y(\pi) = 0. \tag{4.1.1}$$

If the solution is analytic it has a Taylor series

$$y(x) = \sum_{n=0}^{\infty} a_n x^n,$$

and the whole task is to find the coefficients a_n. Substitute into (4.1.1):

$$\frac{d^2}{dx^2}\left(\sum_{n=0}^{\infty} a_n x^n\right) + \left(\sum_{n=0}^{\infty} a_n x^n\right) = 0.$$

Differentiating term by term, the first sum loses its $n = 0, 1$ terms (they are annihilated by the second derivative), so it starts at $n = 2$:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0.$$

Guitar string: the harmonic oscillator by Taylor series (cont.)

To combine the sums we line up the powers of x. Replace $n \mapsto n+2$ in the first sum; its lower limit moves from $n = 2$ to $n = 0$ and the upper limit stays ∞:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}\, x^n + \sum_{n=0}^{\infty} a_n x^n = 0.$$

Now both sums carry x^n, so match coefficients:

$$\sum_{n=0}^{\infty} \Big[(n+2)(n+1)a_{n+2} + a_n\Big] x^n = 0.$$

This holds for all x only if every bracket vanishes:

$$(n+2)(n+1)a_{n+2} + a_n = 0 \quad \Longrightarrow \quad a_{n+2} = -\frac{1}{(n+1)(n+2)}\, a_n.$$

The recurrence skips every other term: a_0 feeds $a_2, a_4, \dots$ and a_1 feeds $a_3, a_5, \dots$.

Resolving the recurrence into $\cos x$ and $\sin x$

Starting from a_0 (even index), each step multiplies by $-1/[(2j-1)(2j)]$:

$$a_2 = -\tfrac{1}{1\cdot 2}a_0 = -\tfrac{1}{2!}a_0, \qquad a_4 = -\tfrac{1}{3\cdot 4}a_2 = \tfrac{1}{4!}a_0, \qquad \ldots$$

Each new step inserts the next two integers into the denominator, so out to a_{2n} the denominator builds up the consecutive product $1\cdot 2\cdot 3\cdots(2n) = (2n)!$ and the sign accumulates n factors of -1:

$$a_{2n} = (-1)^n \frac{1}{(2n)!}\, a_0.$$

Starting from a_1 (odd index), each step multiplies by $-1/[(2j)(2j+1)]$:

$$a_3 = -\tfrac{1}{2\cdot 3}a_1, \qquad a_5 = -\tfrac{1}{4\cdot 5}a_3 = \tfrac{1}{2\cdot 3\cdot 4\cdot 5}a_1 = \tfrac{1}{5!}a_1, \qquad \ldots$$

Here the denominator runs $2\cdot 3\cdots(2n+1) = (2n+1)!$, so

$$a_{2n+1} = (-1)^n \frac{1}{(2n+1)!}\, a_1.$$

Hence the general solution of (4.1.1) is

$$y(x) = a_0 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} + a_1 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

Resolving the recurrence into $\cos x$ and $\sin x$ (cont.)

These two Taylor series are exactly $\cos x$ and $\sin x$:

$$y(x) = a_0 \cos x + a_1 \sin x.$$

The boundary conditions $y(0) = 0$ and $y(\pi) = 0$ force $a_0 = 0$, leaving the string's mode shape $y(x) = a_1 \sin x$.

Drum: Bessel's equation needs a generalized series

Now solve Bessel's equation

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - m^2)\, y = 0. \tag{4.1.2}$$

For the drum m is a nonnegative integer. The equation makes sense for any m, but in this Frobenius derivation we keep the drum case $m = 0, 1, 2, \ldots$. We cannot assume a plain Taylor series: the solution may have a singularity at $x = 0$. So we allow a fractional/negative leading power r (with $a_0 \neq 0$):

$$y(x) = x^r \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+r}. \tag{4.1.3}$$

Substituting (4.1.3) into (4.1.2) and differentiating each series,

$$x^2 \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2} + x \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1} + (x^2 - m^2) \sum_{k=0}^{\infty} a_k x^{k+r} = 0.$$

Because r is still unknown, no term is guaranteed to drop, so we keep every sum starting at $k = 0$. Distributing x^2, x, and $(x^2 - m^2)$:

$$\sum a_k (k+r)(k+r-1) x^{k+r} + \sum a_k (k+r) x^{k+r} + \sum a_k x^{k+r+2} - \sum a_k m^2 x^{k+r} = 0.$$

Drum: Bessel's equation needs a generalized series (cont.)

In the third sum (the only one with x^{k+r+2}) set the new index $j = k+2$, so $k = j-2$ and the term $a_k x^{k+r+2}$ becomes $a_{j-2}x^{j+r}$; renaming j back to k gives $\sum_{k=2}^{\infty} a_{k-2}x^{k+r}$. Extending it down to $k=0$ with the bookkeeping $a_{-2} = a_{-1} = 0$ (those coefficients do not exist), all four sums now share the power x^{k+r} and the same lower limit:

$$\sum_{k=0}^{\infty}\Big(a_k(k+r)(k+r-1) + a_k(k+r) + a_{k-2} - a_k m^2\Big)x^{k+r} = 0.$$

Collect the a_k terms. Since $(k+r)(k+r-1) + (k+r) = (k+r)^2$,

$$\Big[(k+r)^2 - m^2\Big]a_k + a_{k-2} = 0. \tag{4.1.4}$$

At $k=0$, with $a_{-2} = 0$ and $a_0 \neq 0$, the bracket itself must vanish:

$$r^2 - m^2 = 0. \tag{4.1.5}$$

This is the *indicial equation*; it fixes the leading exponent, $r = \pm m$. Take $r = +m$ with $m \geq 0$ so the solution stays finite at $x = 0$.

The Bessel recurrence and $J_m(x)$

With $r = m$, the bracket in (4.1.4) becomes $(k+m)^2 - m^2 = k(k+2m)$, so

$$k(k+2m)\, a_k = -a_{k-2}. \tag{4.1.6}$$

At $k = 1$ this reads $(1+2m)a_1 = 0$. Since $m \geq 0$ (so $1 + 2m \neq 0$), we get $a_1 = 0$, and therefore every odd coefficient vanishes. For the even ones,

$$a_k = \frac{-1}{k(k+2m)}\, a_{k-2}.$$

Iterating from a_0, and writing each new denominator $k(k+2m)$ with $k = 2, 4, \ldots$:

$$\begin{aligned}
a_2 &= \frac{-1}{2(2+2m)} a_0 = \frac{-1}{4(1+m)} a_0, \\
a_4 &= \frac{-1}{4(4+2m)} a_2 = \frac{1}{8 \cdot 4\,(2+m)(1+m)} a_0, \\
a_6 &= \frac{-1}{6(6+2m)} a_4 = -\frac{1}{4^3\, 3!\,(m+3)(m+2)(m+1)} a_0.
\end{aligned}$$

and the odd coefficients $a_3 = a_5 = \cdots = 0$ since $a_1 = 0$. At step j the recurrence contributes a factor $-1/[\,(2j)(2j+2m)\,] = -1/[\,4j(j+m)\,]$. Multiplying these j factors

The Bessel recurrence and $J_m(x)$ (cont.)

collects a $(-1)^j$, a 4^{-j}, a $j! = 1 \cdot 2 \cdots j$ from the first factors, and $(m+1)(m+2)\cdots(m+j) = (m+j)!/m!$ from the second:

$$a_{2j} = (-1)^j \, \frac{m!}{2^{2j}\, j!\, (m+j)!}\, a_0, \qquad a_{2j+1} = 0. \tag{4.1.7}$$

The Bessel recurrence and $J_m(x)$ (cont.)

So the series solution (4.1.3), summing $a_{2j}x^{2j+m}$, is

$$y(x) = a_0\, m! \sum_{j=0}^{\infty} \frac{(-1)^j}{2^{2j}\, j!\,(m+j)!}\, x^{2j+m}.$$

Pull the powers of two into a half-argument: $x^{2j+m}/2^{2j} = 2^m\,(x/2)^{2j+m}$, so

$$y(x) = a_0\, 2^m\, m! \sum_{j=0}^{\infty} \frac{(-1)^j}{j!\,(m+j)!} \left(\frac{x}{2}\right)^{2j+m}.$$

Defining the **Bessel function of the first kind** of order m,

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(m+k)!} \left(\frac{x}{2}\right)^{2k+m}, \tag{4.1.8}$$

the solution reads $y(x) = a_0\, 2^m\, m!\, J_m(x)$.

Fixing the constant. For $m = 0, 1, 2, \ldots,$ the leading coefficient a_0 was free; the standard choice $a_0 = 1/(2^m\, m!)$ cancels the prefactor $2^m\, m!$ and leaves exactly

The Bessel recurrence and $J_m(x)$ (cont.)

$y(x) = J_m(x)$. (This is the conventional normalization, fixing J_m so that its leading term is $(x/2)^m/m!$.)

For an integer order $-m$ ($m > 0$) the method fails: the recurrence $a_k = -a_{k-2}/[\,k(k-2m)\,]$ blows up at $k = 2m$, where the denominator hits zero. Frobenius does not always deliver both solutions; Bessel's second solution waits for §4.3.

What the other Bessel root is trying to say

Bessel's equation contains m^2, so it is often cleaner to write the order as ν and remember that the indicial roots are

$$r = \nu \qquad \text{and} \qquad r = -\nu.$$

The root $r = \nu$ gave the regular solution. With the conventional normalization, and with factorials replaced by gamma functions when ν is not an integer, it is

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu}.$$

If $\nu \notin \mathbb{Z}$, the other root gives the companion Frobenius solution

$$J_{-\nu}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k-\nu+1)} \left(\frac{x}{2}\right)^{2k-\nu}.$$

For noninteger ν, these two leading powers x^ν and $x^{-\nu}$ are different enough to make the solutions independent.

What the other Bessel root is trying to say (cont.)

For order $\nu = \frac{1}{2}$, the root $r = -\frac{1}{2}$ gives the companion half-order solution in its simplest form. Choose that root. Then

$$\left[(k - \tfrac{1}{2})^2 - \tfrac{1}{4}\right] a_k + a_{k-2} = 0 \quad \Longleftrightarrow \quad k(k-1)a_k = -a_{k-2}.$$

The $k = 1$ equation is $0 \cdot a_1 = 0$, so a_1 is free instead of being forced to vanish. The two chains are

$$a_{2j} = (-1)^j \frac{a_0}{(2j)!}, \qquad a_{2j+1} = (-1)^j \frac{a_1}{(2j+1)!}.$$

Thus

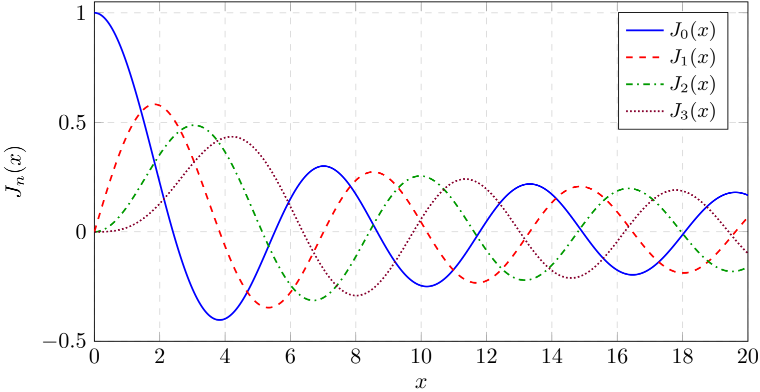
$$y(x) = x^{-1/2}\big(a_0 \cos x + a_1 \sin x\big).$$

After standard normalization,

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x, \qquad J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x.$$

For positive integer order, the negative-root branch is the one that breaks; the missing independent solution is the singular Y_m constructed later by reduction of order.

The Bessel functions of the first kind $J_0, \ldots, J_3$



Near the origin, $J_0(0) = 1$ while $J_m(x)$ for $m \geq 1$ first behaves like a constant multiple of x^m. Farther out the Bessel functions oscillate with a slowly decaying amplitude. The zeros of J_m are the allowed radial constants k that clamp the drum's rim.

The method of Frobenius: the recipe

To solve $p_0(x)y'' + p_1(x)y' + p_2(x)y = 0$:

1. Substitute the generalized series $y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$.
2. Shift the indices so all powers of x are the same.
3. Combine the matching coefficients together.
4. Solve the *indicial equation* (the $k = 0$ coefficient) to determine r.
5. Develop a recursion formula starting with a_0.

The leading exponent r comes from the lowest power of x; once r is fixed, the recurrence generates all higher coefficients from a_0 (and, when the recurrence allows, a_1).

Example: Airy's equation by series

Airy's equation

$$y'' - xy = 0 \tag{4.1.9}$$

arises for a quantum particle in a constant force (linear potential) and in the diffraction of light. Its behavior is already readable off the equation: for $x < 0$ it looks like $y'' + \lambda^2 y = 0$ (oscillatory, $\sin \lambda x, \cos \lambda x$), and for $x > 0$ like $y'' - \lambda^2 y = 0$ (exponential, $e^{\pm \lambda x}$).
Take $y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$, so $y'' = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1)x^{k+r-2}$, and (4.1.9) becomes

$$\sum_{k=0}^{\infty} a_k (k+r)(k+r-1)x^{k+r-2} - \sum_{k=0}^{\infty} a_k x^{k+r+1} = 0.$$

Align powers: in the second sum the power is x^{k+r+1}, and setting the new index $j = k+3$ (so $k = j-3$) turns $a_k x^{k+r+1}$ into $a_{j-3} x^{j+r-2}$; renaming $j \to k$ and using $a_{-3} = a_{-2} = a_{-1} = 0$ to start at $k = 0$,

$$\sum_{k=0}^{\infty} \Big(a_k (k+r)(k+r-1) - a_{k-3} \Big) x^{k+r-2} = 0.$$

Example: Airy's equation by series (cont.)

Hence the coefficient of each power vanishes:

$$a_k(k+r)(k+r-1) - a_{k-3} = 0.$$

The $k=0$ term is the indicial equation $r(r-1)=0$, so $r=0$ or $r=1$. Here it does not matter which we take; both give a Taylor series ($x=0$ is an ordinary point). Take $r=0$. Then $a_k(k)(k-1) - a_{k-3} = 0$; replacing k by $k+3$ (so that $(k+3)(k+3-1) = (k+3)(k+2)$) puts the recurrence in forward form:

$$k(k-1)a_k - a_{k-3} = 0 \quad \Longrightarrow \quad a_{k+3} = \frac{1}{(k+3)(k+2)}\, a_k.$$

Example: Airy's equation by series (cont.)

The index jumps by 3, so a_0, a_1, a_2 seed three residue-class chains. At $k = 2$ the relation reads $2(1)a_2 - a_{-1} = 0$, and with $a_{-1} = 0$ this forces $a_2 = 0$, killing the a_{3k+2} chain. From a_0 (each step multiplies by $1/[(3j)(3j-1)]$):

$$\begin{aligned} a_3 &= \frac{1}{3 \cdot 2} a_0, \\ a_6 &= \frac{1}{6 \cdot 5} a_3 = \frac{1}{6 \cdot 5 \cdot 3 \cdot 2} a_0, \\ a_9 &= \frac{1}{(9 \cdot 8)(6 \cdot 5)(3 \cdot 2)} a_0, \\ a_{3k} &= \frac{(3k-2) \cdot (3k-5) \cdots 1}{(3k)!}\, a_0. \end{aligned}$$

The numerator is a product stepping down by 3. Writing the *multifactorial* $n!^{(j)} = n \cdot (n-j)!^{(j)}$ with $n!^{(j)} = 1$ for $-j < n \le 0$, this is $(3k-2)!!!$ (triple factorial), so

$$a_{3k} = \frac{(3k-2)!!!}{(3k)!}\, a_0.$$

Example: Airy's equation by series (cont.)

The a_1 chain works the same way, shifted by one power:

$$\begin{aligned} a_4 &= \frac{1}{4 \cdot 3} a_1, \\ a_7 &= \frac{1}{7 \cdot 6} a_4 = \frac{1}{7 \cdot 6 \cdot 4 \cdot 3} a_1, \\ a_{10} &= \frac{8!!!}{10!} a_1, \\ a_{3k+1} &= \frac{(3k-1)!!!}{(3k+1)!}\, a_1. \end{aligned}$$

$$\begin{aligned} y(x) = a_0 \sum_{k=0}^{\infty} \frac{(3k-2)!!!}{(3k)!}\, x^{3k} \\ + a_1 \sum_{k=0}^{\infty} \frac{(3k-1)!!!}{(3k+1)!}\, x^{3k+1}. \end{aligned} \tag{4.1.10}$$

Mathematical detour: Airy via Fourier transform

Airy's equation can also be solved by transforming it. The calculation below is a formal Fourier-transform derivation: one can justify the integrations by parts and inversion by temporarily damping the oscillatory integrals and then taking the damping to 0^+, or by working in the distribution sense. With

$$\hat{y}(t) = \mathrm{F}\{y(x)\}(t) = \int_{-\infty}^{\infty} y(x)\, e^{-itx}\, dx, \qquad y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{y}(t)\, e^{itx}\, dt.$$

Integration by parts gives the derivative rule

$$\mathrm{F}\{y'\} = it\, \mathrm{F}\{y\}, \qquad \text{repeating once:} \qquad \mathrm{F}\{y''\} = -t^2\, \mathrm{F}\{y\}.$$

For multiplication by x, differentiate $\hat{y}(t) = \int y\, e^{-itx} dx$ under the integral:

$$\hat{y}\,'(t) = \int y\,(-ix)\, e^{-itx}\, dx = -i\, \mathrm{F}\{xy\} \quad \Longrightarrow \quad \mathrm{F}\{xy\} = +i\, \big(\mathrm{F}\{y\}\big)'.$$

Mathematical detour: Airy via Fourier transform (cont.)

Transforming $y'' - xy = 0$,

$$\mathrm{F}\{y'' - xy\} = \mathrm{F}\{y''\} - \mathrm{F}\{xy\} = -t^2\hat{y} - i\,\hat{y}\,' = 0.$$

This is separable: $-i\hat{y}\,' = t^2\hat{y}$, i.e.

$$\frac{\hat{y}\,'}{\hat{y}} = it^2 \quad \Longrightarrow \quad \hat{y} = c\,e^{it^3/3}.$$

(Using $1/(-i) = i$.) Inverting,

$$y(x) = \frac{c}{2\pi}\int_{-\infty}^{\infty} e^{it^3/3}\,e^{itx}\,dt = \frac{c}{2\pi}\int_{-\infty}^{\infty} e^{i(t^3/3+tx)}\,dt = \frac{c}{\pi}\int_0^{\infty}\cos\!\left(\frac{t^3}{3} + tx\right)dt,$$

the last step keeping the even (cosine) part since the odd part integrates to zero. We name

$$Ai(x) = \frac{1}{\pi}\int_0^{\infty}\cos\!\left(\frac{t^3}{3} + xt\right)dt, \tag{4.1.11}$$

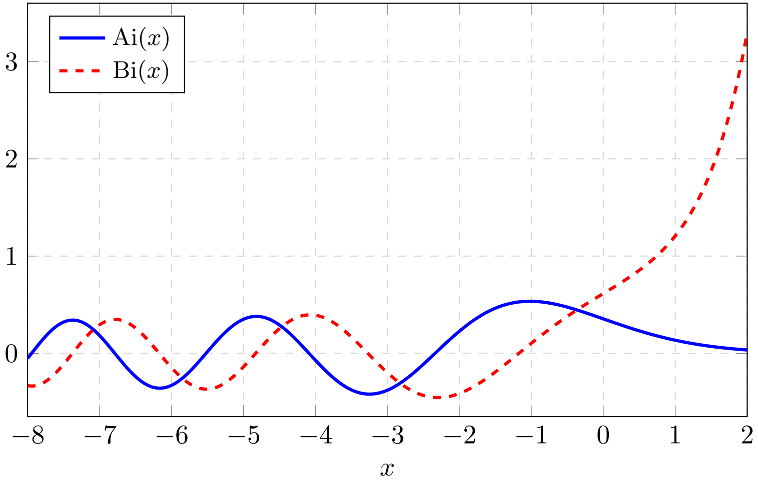
the **Airy function of the first kind**, and the second kind

$$Bi(x) = \frac{1}{\pi}\int_0^{\infty}\left[\exp\!\left(-\frac{t^3}{3} + xt\right) + \sin\!\left(\frac{t^3}{3} + xt\right)\right]dt. \tag{4.1.12}$$

Mathematical detour: Airy via Fourier transform (cont.)

The real-line Fourier transform selected one particular solution, Ai; Bi is the other standard independent Airy solution and is introduced here for comparison. A known value (from the gamma function) is $Ai(0) = \left(3^{2/3}\,\Gamma(2/3)\right)^{-1}$.

The Airy functions *Ai* and *Bi*



For $x < 0$ both standard Airy solutions oscillate. For $x > 0$, $Ai(x)$ decays while $Bi(x)$ grows, so boundary conditions usually decide which combination is physically allowed.

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Classifying the point x_0

For $y'' + P(x)y' + Q(x)y = 0$, classify the point x_0 by the coefficients P and Q:

- x_0 is an **ordinary point** if P and Q are analytic at x_0.
- x_0 is a **regular singular point** if x_0 is singular and $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0.
- Otherwise x_0 is an **irregular singular point**.

Classifying the point x_0 (cont.)

The behavior of P and Q predicts the outcome.

1. If both $P(x)$ and $Q(x)$ are analytic near x_0 (in particular, no pole there), then x_0 is an *ordinary point*. The equation has *two* distinct series solutions; the radius of convergence is at least the smaller of the radii of P and Q.
2. **Fuchs' theorem.** If x_0 is singular, P has at most a simple pole, and Q has at most a double pole at x_0—i.e. $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0—then x_0 is a *regular singular point*, and there is at least one series solution

$$y = \sum_{k=0}^{\infty} a_k (x - x_0)^{k+r}.$$

   If the two roots of the indicial equation differ by a noninteger, there are two independent Frobenius solutions. If the roots coincide or differ by an integer, at least one Frobenius solution is guaranteed; when the roots are distinct, it is usually the larger-root solution.
3. Otherwise x_0 is an *irregular singular point*, and the equation may have no series solution at all.

Examples: reading off P and Q at $x = 0$

Airy, $y'' - xy = 0$**.** Here $P(x) = 0$ and $Q(x) = -x$ are both analytic, so $x = 0$ is an *ordinary point*: two Taylor-series solutions, as we found.

$x^4y'' + 2x^3y' + y = 0$**.** Divide by x^4: $y'' + \frac{2}{x}y' + \frac{1}{x^4}y = 0$, so $P(x) = 2/x$ is a *simple* pole (order 1) and $Q(x) = 1/x^4$ has a pole of order 4. The order-4 pole exceeds the double-pole limit, so $x = 0$ is an *irregular singular point*: no Frobenius power-series solution at the origin. Indeed the true solution is

$$y(x) = c_1 \cos\left(\frac{1}{x}\right) + c_2 \sin\left(\frac{1}{x}\right),$$

which has an essential singularity at the origin.

Examples: reading off P and Q at $x = 0$ (cont.)

$x^2y'' - 6y = 0$. Divide by x^2: $y'' + 0 \cdot y' - \frac{6}{x^2}y = 0$, so $x = 0$ is singular, $P(x) = 0$ is analytic, and $Q(x) = -6/x^2$ has a second-order pole. This meets Fuchs' theorem, so $x = 0$ is a *regular singular point* and we get at least one solution. Substituting $y = \sum a_k x^{k+r}$ into $x^2y'' - 6y = 0$:

$$\sum_{k=0}^{\infty} \Big[(k+r)(k+r-1) - 6\Big] a_k x^{k+r} = 0 \quad \Longrightarrow \quad \Big[(k+r)(k+r-1) - 6\Big] a_k = 0.$$

At $k = 0$ the indicial equation is

$$r(r-1) - 6 = 0 \quad \Longrightarrow \quad r = 3 \text{ or } r = -2.$$

This is *not* a recurrence (no a_{k-2} couples the coefficients): from a_0 we cannot generate $a_1, a_2, \ldots$. But both indicial roots give a solution, $y = x^3$ and $y = x^{-2}$, so

$$y(x) = c_1 x^3 + \frac{c_2}{x^2}.$$

Examples: reading off P and Q at $x = 0$ (cont.)

$x^3 y'' - 6y = 0$. Now $Q(x) = -6/x^3$ has a *third*-order pole, beyond Fuchs' limit. We should not expect a Frobenius power series at the origin. Substituting $y = \sum a_k x^{k+r}$ into $x^3 y'' - 6y = 0$ gives

$$\sum_{k=0}^{\infty}(k+r)(k+r-1)a_k x^{k+r+1} - 6\sum_{k=0}^{\infty} a_k x^{k+r} = 0.$$

The lowest power x^r comes only from the second sum, so its coefficient is the indicial equation $-6a_0 = 0$. Since $a_0 \neq 0$ this is impossible: there are no Frobenius solutions of the assumed form.

Bessel, $x^2 y'' + xy' + (x^2 - m^2)y = 0$. Divide by x^2:

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0.$$

So $P(x) = 1/x$ has a simple pole and Q has at most a double pole (a genuine double pole when $m \neq 0$; no pole in Q when $m = 0$): Fuchs' theorem guarantees at least one Frobenius solution. For integer $m \geq 0$, the $r = m$ solution gives J_m; the second independent solution is not a pure Frobenius series. If the indicial roots differ by a noninteger, there are two Frobenius solutions.

Summary: when Frobenius works

For $y'' + P(x)y' + Q(x)y = 0$ at the point x_0:

- P, Q both analytic $\Rightarrow$ **two** analytic (series) solutions centered at x_0.
- P at most a **simple** pole and Q at most a **double** pole (regular singular point) $\Rightarrow$ **at least one** series solution.
- Otherwise (irregular singular point) there is no general Frobenius guarantee; the method may fail.

When only one solution is guaranteed, the second must be found another way — the subject of the next section.

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Reduction of order: building the second solution

Frobenius gave only one solution to Bessel's equation, but a second-order equation has *two* independent solutions, just as $y'' + y = 0$ has $a_0 \cos x$ and $a_1 \sin x$. Suppose $y(x)$ and $v(x)$ both solve $y'' + Py' + Qy = 0$:

$$(1) \quad y'' + Py' + Qy = 0, \qquad (2) \quad v'' + Pv' + Qv = 0.$$

Form $v \cdot (1) - y \cdot (2)$; the two Q terms give $Q(vy - yv) = 0$ and drop out, leaving

$$(vy'' - yv'') + P\,(vy' - yv') = 0.$$

Notice that $(vy' - yv')' = v'y' + vy'' - y'v' - yv'' = vy'' - yv''$ (the $v'y'$ and $y'v'$ cross terms cancel). Writing $W = vy' - yv'$ (the Wronskian-type combination), the first group is exactly W':

$$W' + P\,W = 0.$$

Reduction of order: building the second solution (cont.)

This is first order. Multiply by the integrating factor $\mu(x) = \exp\left(\int_a^x P(t)\,dt\right)$, whose derivative is $\mu' = P\mu$. Then $\mu W' + P\mu W = \mu W' + \mu' W = (\mu W)'$, so $\mu(W' + PW) = 0$ collapses to a total derivative:

$$\frac{d}{dx}\left[\exp\left(\int_a^x P(t)\,dt\right)(vy' - yv')\right] = 0.$$

Hence the bracket is a constant B:

$$\exp\left(\int_a^x P(t)\,dt\right)(vy' - yv') = B.$$

On an interval where v is nonzero, solve for the combination and divide by v^2:

$$\frac{vy' - yv'}{v^2} = \frac{B\,\exp\left(-\int_a^x P(t)\,dt\right)}{v^2}.$$

Reduction of order: building the second solution (cont.)

The left side is an exact derivative, $\dfrac{d}{dx}\left(\dfrac{y}{v}\right)$. Integrating gives $y/v = C + B\int^x \exp\!\big(-\int_a^s P(t)\,dt\big)/v^2(s)\,ds$. The term $Cv(x)$ is just the solution we already know, so set $C = 0$ to isolate the *second solution* from the first one v:

$$y_2(x) = B\,v(x)\int^x \frac{\exp\!\big(-\int_a^s P(t)\,dt\big)}{v^2(s)}\,ds. \tag{4.3.1}$$

Two examples of reduction of order

Harmonic oscillator, $y'' + y = 0$**.** We already know $\cos x$ and $\sin x$, but let us recover $\sin x$ from $\cos x$ using (4.3.1). Work on any interval avoiding the zeros of $\cos x$, for example $|x| < \pi/2$. Here $P(x) = 0$ and $v(x) = \cos x$, so the exponential factor is 1:

$$y(x) = B \cos x \int_0^x \frac{1}{\cos^2 s}\, ds = B \cos x \,\tan x = B \sin x.$$

The reduction formula reproduces the expected partner.

Two examples of reduction of order (cont.)

Bessel's equation, $x^2y'' + xy' + (x^2 - m^2)y = 0$. Here $P(x) = 1/x$ and the known solution is $v(x) = J_m(x)$. The exponential factor is

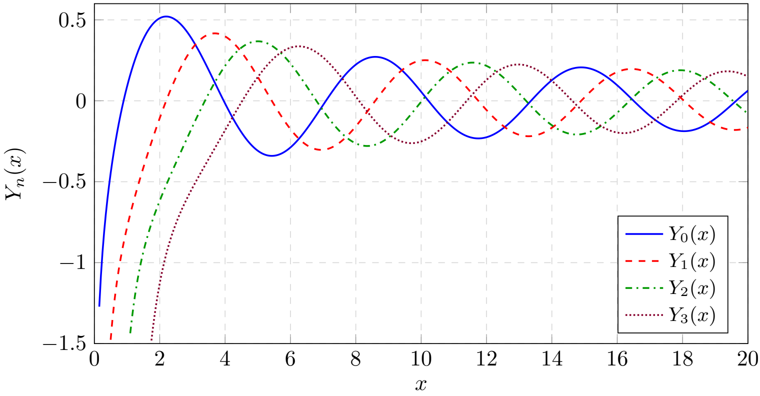
$$\exp\Bigl(-\int_1^x t^{-1}\,dt\Bigr) = \exp(-\ln x) = x^{-1}.$$

Substituting into (4.3.1),

$$y(x) = B\,J_m(x)\int_{x_*}^{x} \frac{1}{s\,J_m^2(s)}\,ds, \tag{4.3.2}$$

where $x_* > 0$ and the interval avoids zeros of J_m. This gives a second independent solution. After the conventional normalization and after adding any harmless multiple of J_m, this solution is the **Bessel function of the second kind**, denoted $Y_m(x)$. For $m > 0$, $J_m(0) = 0$ and the integrand is strongly singular; for $m = 0$, $J_0(0) = 1$ and the $1/s$ factor gives the logarithmic singularity. Thus Y_m is singular at the origin — the second, independent solution that Frobenius alone could not reach.

The Bessel functions of the second kind $Y_0, \ldots, Y_3$



Unlike J_m, each Y_m plunges to $-\infty$ as $x \to 0^+$: the second Bessel solution is singular at the center, so it is excluded for a drum (finite there) but needed on annular domains.

Chapter summary

- The **method of Frobenius** solves linear second-order homogeneous ODEs by a generalized power series $y = \sum_k a_k x^{k+r}$.
- The lowest power gives the **indicial equation** for r; matching higher powers gives a **recurrence** for the a_k.
- The harmonic oscillator yields $\cos x, \sin x$; Bessel's equation yields the **Bessel function of the first kind** $J_m(x) = \sum_k \frac{(-1)^k}{k!\,(m+k)!}(x/2)^{2k+m}$; Airy's equation yields multifactorial series and the integral $Ai(x), Bi(x)$.
- The other Bessel root gives $J_{-\nu}$ for noninteger order; for integer order it does not provide a new independent solution, which is why Y_m is needed.
- **Fuchs' theorem**: ordinary point $\Rightarrow$ two series solutions; regular singular point (P simple pole, Q double pole) $\Rightarrow$ at least one; irregular singular point $\Rightarrow$ possibly none.
- **Reduction of order** recovers the missing second solution: $y = B\,v \int \exp(-\int P)\,/\,v^2\,dx$, giving e.g. the Bessel function of the second kind Y_m.